Aditya Kumar

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Education

Dayananda Sagar College of Engineering Bengaluru, IN
Bachelor of Engineering in Computer Science *Sep. 2025 – Sep. 2029*

Skills

Languages: Golang ,Python ,JavaScript ,TypeScript
Database: PostgreSQL ,MongoDB ,Redis
Frameworks: Next.js ,Gin ,React ,Node.js ,Express
Developer Tools: Github Actions ,Docker ,Kubernetes ,Postman ,Linux
Libraries : WebRTC, Socket.IO ,React-map-gl ,Cobra ,Bubble Tea

Projects

BlackBox | *Golang, Bubble Tea(TUI)* GitHub

- Built a fast command line HTTP download manager in Go to reliably fetch and save files directly from the terminal.
- Created a snappy and interactive Terminal UI using Bubble Tea and Lip Gloss, allowing users to easily queue and monitor multiple downloads.
- Managed background downloads using goroutines and strictly applied mutex locks to prevent data race conditions between the UI and downloader threads.
- Tracked and displayed live download speeds, progress bars, and ETAs.

Droppa | *WebRTC, WebSockets, Next.js* GitHub

- Built a P2P file transfer app using WebRTC DataChannels, which enables direct device-to-device file transfer with no upload/download size limits.
- Implemented chunked file transfer over encrypted RTCDataChannel with backpressure handling via bufferedAmountLow events, preventing high memory usage and browser crashes when transferring multi GB files across unreliable connections.
- Built a Socket.io signaling server to orchestrate SDP offer/answer exchange and ICE candidate trickle, enabling peers to establish connection.
- Improved user experience with dynamic QR codes tied to unique room IDs, reducing session setup to a single scan.

Armoracrypt | *Golang, Cobra, Dropbox-API* GitHub

- ArmoraCrypt is a Golang-based CLI tool to facilitate local AES-256 encryption/decryption of files and folders, enabling secure uploads/downloads to cloud storage while safeguarding against provider surveillance and unauthorized access.
- Integrated zipping and unzipping functionality to handle entire folders, streamlining the encryption process for complex directory structures.
- Designed a master-key initialization command for generating and managing local encryption keys, ensuring a secure and self-contained key management system.
- Implemented upload and download features for encrypted files to cloud storage, allowing users to safely store and retrieve sensitive data.

Litspots | *Next.js, MongoDB, React-map-gl* GitHub

- Built a full-stack real-time social discovery web app enabling users to find, join, and host live hangouts nearby.
- Reduced database load by caching recent API responses and serving stale data to prevent excessive queries.
- Integrated interactive maps with MapLibre GL and react-map-gl to display live hangout locations.
- Used a geocoding API to convert place names into geographic coordinates for accurate map-based discovery.
- Implemented backend APIs and custom authentication using Next.js API Routes, MongoDB, JWT, and bcryptjs.

Achievements

ACM Summer School @IIT Hyderabad

Runner up in beyond Abstraction (ETH india) hackathon - Awarded $1000